

5.18 (a) From Problem 5.17 (a), we know that the image current is $\frac{\mu - \mu_0}{\mu + \mu_0} I \equiv I'$, located at $z = -d$, and has the same direction as the original current, if it is located at $z = d$. From Section 5.5, the vector potential from the image current is given by Eq. (5.35),

$$A_\phi(r, \theta) = \frac{\mu_0}{4\pi} \frac{4I'a}{\sqrt{a^2 + r^2 + 2ars\sin\theta}} \left[\frac{(z-k^2)K(k) - zE(k)}{k^2} \right],$$

with $k^2 = 4ars\sin\theta / (a^2 + r^2 + 2ars\sin\theta)$. The original current loop is located at

$r = \sqrt{a^2 + (2d)^2}$, and $\sin\theta = a/r$. Therefore, the vector potential along the current loop is

$$A_\phi(r, \theta) = \frac{\mu_0 I'}{2\pi} \frac{a}{\sqrt{a^2 + d^2}} \left[\frac{(z-k^2)K(k) - zE(k)}{k^2} \right] = \frac{\mu_0 I'}{2\pi} \left[\left(\frac{2}{k} - k \right) K(k) - \frac{2}{k} E(k) \right],$$

as $k = a / \sqrt{a^2 + d^2}$. Now, we can see that the vector potential on the original current loop is a constant and is in the same direction of the original current, the interaction energy is

$$W = \int \vec{j} \cdot \vec{A} d\ell = \mu_0 I I' a \left[\left(\frac{2}{k} - k \right) K(k) - \frac{2}{k} E(k) \right].$$

To find the force acting on the loop, take the derivative with respect to $2d$, we will have

$$F = \frac{1}{2} \frac{\partial W}{\partial d} = \frac{\mu_0 I I' a}{2} \left[\left(-\frac{2}{k^2} - 1 \right) K(k) + \left(\frac{2}{k} - k \right) \frac{dK(k)}{dk} + \frac{2}{k^2} E(k) - \frac{2}{k} \frac{dE(k)}{dk} \right] \frac{\partial k}{\partial d}.$$

Since $\frac{dK(k)}{dk} = \frac{E(k)}{k(1-k^2)} - \frac{K(k)}{k}$, $\frac{dE(k)}{dk} = \frac{E(k)}{k} - \frac{K(k)}{k}$, and $\frac{\partial k}{\partial d} = -\frac{ad}{(a^2 + d^2)^{3/2}}$,

we have

$$F = -\frac{\mu_0 I I' a}{2} \left[-\frac{2}{k^2} K(k) + \frac{1}{1-k^2} \left(\frac{2}{k^2} - 1 \right) E(k) \right] \frac{ad}{(a^2 + d^2)^{3/2}}.$$

(b) Skip.

c) For $d \gg a$, $k \ll 1$ we can use the Maclaurin series of $K(k)$ and $E(k)$,

$$K(k) = \frac{\pi}{2} \left(1 + \frac{k^2}{4} + \frac{9}{64} k^4 \right), \quad E(k) = \frac{\pi}{2} \left(1 - \frac{k^2}{4} - \frac{9}{64} k^4 \right),$$

to write the force as

$$\begin{aligned} F &= - \frac{\mu_0 I I' a}{2} \left[- \frac{2}{k^2} \cdot \frac{\pi}{2} \left(1 + \frac{k^2}{4} + \frac{9}{64} k^4 \right) + \left(\frac{2}{k^2} + \frac{1}{1-k^2} \right) \cdot \frac{\pi}{2} \left(1 - \frac{k^2}{4} - \frac{9}{64} k^4 \right) \right] \frac{ad}{(a^2+d^2)^{3/2}} \\ &= - \frac{\mu_0 I I' a}{2} \left[- \frac{\pi}{2} \frac{2}{k^2} \left(\frac{k^2}{2} + \frac{3}{16} k^4 \right) + \frac{\pi}{2} \frac{1}{1-k^2} \left(1 - k^2 + \frac{3}{4} k^2 - \frac{3}{4} k^4 + \frac{45}{64} k^4 \right) \right] \frac{ad}{(a^2+d^2)^{3/2}} \\ &= - \frac{\mu_0 I I' a \pi}{4} \left[-1 - \frac{3}{8} k^2 + 1 + \frac{3}{4} k^2 + o(k^2) \right] \frac{ad}{(a^2+d^2)^{3/2}} \\ &= - \frac{\mu_0 I I' a \pi}{4} \frac{3k^2}{8} \frac{ad}{(a^2+d^2)^{3/2}} \end{aligned}$$

Since $k^2 \sim a^2/d^2$, we finally have

$$F = - \frac{\mu_0}{2\pi} (I \pi a^2) (I' \pi a^2) \frac{3}{(2d)^4}$$

For a magnetic dipole $\vec{m}$, the magnetic field generated by it is

$$\vec{B} = \frac{\mu_0}{4\pi} \frac{3\vec{n}(\vec{n} \cdot \vec{m}) - \vec{m}}{r^3}$$

The interaction energy between this magnetic dipole and another magnetic dipole $\vec{m}$ is therefore

$$W = \vec{m} \cdot \vec{B} = \frac{\mu_0}{4\pi} \frac{3(\vec{n} \cdot \vec{m}_1)(\vec{n} \cdot \vec{m}_2) - \vec{m}_1 \cdot \vec{m}_2}{r^3}$$

If they are in the same direction, there will be an attractive force between them, directing toward each other, given by

$$F = \frac{\partial W}{\partial r} = - \frac{\mu_0}{4\pi} \frac{3[3(\vec{n} \cdot \vec{m}_1)(\vec{n} \cdot \vec{m}_2) - \vec{m}_1 \cdot \vec{m}_2]}{r^4}$$

In part (a), $\vec{m}_1 = I \pi a^2 \hat{z}$, $\vec{m}_2 = I' \pi a^2 \hat{z}$, and $\vec{n} = \hat{z}$, $r = 2d$, the force is

$$F = - \frac{\mu_0}{4\pi} \frac{3(I \pi a^2)(I' \pi a^2)}{8d^4}$$

which agrees the limiting case of part (a).